Summary

1) For MBS, volatility risk can vary significantly across different systems even when they are using the same prepayment model and interest rate process. These differences come from whether one incorporates volatility as part of mortgage rate process.
2) First we describe in details on how volatility enters into the mortgage rate process. Typically, the mortgage rate process starts with a current coupon which is derived from TBA prices. If the interest rate curve remains unchanged but volatility increases, one would expect TBA prices to fall which will result in an increase of the current coupon. Some examples of systems that incorporate volatility into the mortgage rate process include Yield Book and Bloomberg
3) If volatility is not part of the mortgage rate process, then one would need to account for its impact on current coupon in order to have an accurate assessment of volatility risk. In other words, given a one unit of volatility increase, we want to determine it impact on mortgage rate/current coupon.
4) Recall that the Cello mortgage rate process consists of a current coupon process and a primary vs. secondary spread model. The current coupon process is a constant spread over the linear combination of two, five, ten and thirty year rates. From now on, we will assume the volatility unit to be basis point, ignore the primary vs secondary spread model and focus on the impact of volatility on current coupon yield and hence the current coupon spread since we assume the curve remains unchanged.
5) Starting with the theoretical CMM102 collateral, we want to compute its Vol dv01 X and CC dv01 Y. This is estimated by computing the Vol dv01 and CC dv01 of the TBA coupons surrounding CMM 102 before applying linear interpolation
6) We also need to calculate the price impact of one basis point change in current coupon yield which is also done for our C3 risk calculation. Recall that we first estimate the up and down 25 basis point prices of the CMM 102 coupon via linear interpolating from the TBA prices. The differences between the two prices normalized by 50 basis points give the required result which we will denote by Z.
7) Now suppose a is the impact on the current coupon yield based on one basis point increase in volatility. We can now compute the price impact on the theoretical CMM102 from 5) and 6) separately resulting in the following equation:

   $$aZ = X + aY$$

   Once we solve for the unknown a from the linear equation above, we can compute the adjusted vol risk of any MBS as:

   $$\text{Vol dv01} + a^* \text{ CC dv01}$$

   whenever volatility is not part of the mortgage rate process.